# Datasets and Sources

*Dataset register, links, processing lineage and attribution notes*

**STAT7006A | Spatial Statistics Dashboard | Final bright-theme candidate**

## 1. Dataset inventory

| File or dataset | Purpose | Availability |
|---|---|---|
| **NASA FIRMS VIIRS NOAA-21 (J2V-C2) archive** | Original satellite active-fire detections; 2024–2025. | Included in source_archives as original downloaded ZIP. |
| **Cleaned vegetation-detection CSV** | Intermediate filtered fire records for provenance/reference. | Included in source_archives as cleaned ZIP. |
| **data/fires_final.rds** | Prepared sf points with province, year, season and month/date for the app. | REQUIRED to run; NOT in this ZIP. Copy from existing project. |
| **data/provinces.rds** | Prepared South African province sf polygons; analysis windows and mapping. | REQUIRED to run; NOT in this ZIP. Copy from existing project. |

## 2. Official data-source links

| Resource | Link / citation guidance |
|---|---|
| **NASA FIRMS homepage** | https://firms.modaps.eosdis.nasa.gov/ |
| **NASA FIRMS archive download** | https://firms.modaps.eosdis.nasa.gov/download/ |
| **NASA FIRMS field definitions** | https://firms.modaps.eosdis.nasa.gov/download/Readme.txt |
| **Provincial polygons: original source** | NOT VERIFIED from available materials. Ask the person who prepared provinces.rds for the provider, version and license. |
| **Alternative boundaries, NOT confirmed used** | https://www.geoboundaries.org/ ; https://gadm.org/ |

The exact NASA FIRMS archive filename is DL_FIRE_J2V-C2_828479.zip; its CSV is fire_archive_J2V-C2_828479.csv. The intermediate vegetation-detection archive is FIREWATCH_SA_NOAA21_2024_2025_vegetation_detections.zip. These names document the project's source lineage but do not by themselves specify the original download date or a citable DOI.

## 3. Reported data-preparation flow

1. Downloaded the VIIRS NOAA-21 active-fire archive for South Africa covering 2024 and 2025.
2. Filtered detections to the archive's presumed vegetation-fire category (type == 0).
3. Assigned fire points spatially to the nine South African provincial polygons.
4. Derived or retained year, month and season attributes.
5. Saved the spatial point data as fires_final.rds and provincial polygons as provinces.rds for use in Shiny.

| Stage | Records | Meaning |
|---|---|---|

| | | |
|---|---|---|
| **Original archive** | 386,424 | Unfiltered VIIRS NOAA-21 records, 2024–2025. |
| **Vegetation category after type filter** | 340,776 | Presumed vegetation-fire detections before final spatial assignment. |
| **Final prepared app data** | 340,156 | Assigned to one of nine South African provinces. |
| **Final 2024 subset** | 216,753 | Year 2024 records in app dataset. |
| **Final 2025 subset** | 123,403 | Year 2025 records in app dataset. |

Counts are based on the established team workflow and previous project checks; they are not freshly recalculated from the missing RDS files in this handover. The difference between the filtered and final counts should be checked against the original preparation script before making a detailed claim about excluded records.

## 4. Variables used by the app

| Field | Expected form | Use |
|---|---|---|
| **geometry** | sf POINT | Maps, spatial projections and point-pattern analysis. |
| **province** | character | Province filter, grouped summaries, analysis study window. |
| **year** | 2024 or 2025 | Year filter and seasonal comparison. |
| **season** | Summer / Autumn / Winter / Spring | Season filter and seasonal summaries. |
| **month or acq_date** | 1–12 or valid acquisition date | Monthly charts and seasonal grouping. |
| **province geometry** | sf MULTIPOLYGON/POLYGON | Map boundaries and analysis windows. |

## 5. Dataset interpretation and limits

- The data measure satellite-detected thermal anomalies, not confirmed individual fires or hectares burned.
- The same fire may appear multiple times across observations or pixels; detections are not independent fire events.
- Cloud, smoke, observation timing, sensor characteristics and geographic conditions affect detectability.
- Counts differ by province area and burnable land cover; raw counts are not normalized fire-risk estimates.
- Spatial clustering can reflect vegetation, climate, topography, land use and observation effects, not necessarily interaction between fire events.
- The project compares calendar-year seasonal counts; summer is not treated as a December-to-February season spanning two years.

## 6. Suggested attribution for report or slides

“Active-fire detection data were obtained from NASA’s Fire Information for Resource Management System (FIRMS), using the VIIRS NOAA-21 (J2V-C2) product for 2024–2025. Records were filtered and assigned to South African provincial polygons to prepare the dashboard dataset.”

Complete the provincial-boundary attribution only after confirming the original provider and version. Do not cite geoBoundaries or GADM as the actual source merely because those are possible download sites.

## 7. Reproducibility checklist

6. Confirm the precise boundary dataset provider, edition, license and download date.
7. Copy the original data-preparation script (01_prepare_data.R) into the project if it is available; it is not included in this handover.
8. Confirm RDS field names and coordinate reference systems from the working project.
9. Keep the original source ZIPs unchanged for provenance.
10. Do not replace the working NOAA-21 data with a teammate’s S-NPP or another VIIRS product without explicit methodological review.